

10. Signal Scaling for Overflow Prevention

Simulate a fixed-point FIR filter with and without signal scaling and evaluate its effectiveness in preventing arithmetic overflow. (25) (BL3) (WK4)

Given Data:

- FIR Filter Length = 12 taps
- Sum of Coefficients = 2.5
- Input Amplitude = ± 1
- Word Length = 8 bits

Tasks:

1. Determine the required scaling factor.
2. Simulate the filter without scaling.
3. Simulate the filter with scaling.
4. Compare output waveforms.
5. Identify overflow occurrences.
6. Demonstrate the effectiveness of scaling.

Simulation Objectives

1. Objective 1: Scaling Factor Determination & Filter Design

Calculate the theoretical scaling factor S using the L_1 -norm bound ($S = 1/\sum |h[n]|$) to prevent output saturation in an 8-bit fixed-point architecture, and visualize the 12-tap FIR impulse response.

2. Objective 2: Unscaled Fixed-Point FIR Filter Simulation & Overflow Detection

Simulate the 12-tap FIR filter without input scaling under dynamic input (± 1), bound the output to signed 8-bit limits ($[-1, 0.9922]$), and plot the unscaled output showing arithmetic overflow/clipping events.

3. Objective 3: Scaled Fixed-Point FIR Filter Simulation

Apply the scaling factor $S = 0.4$ to the input signal prior to filtering and verify that the output remains within dynamic range limits without clipping.

4. Objective 4: Comparative Output Analysis & Scaling Effectiveness Verification

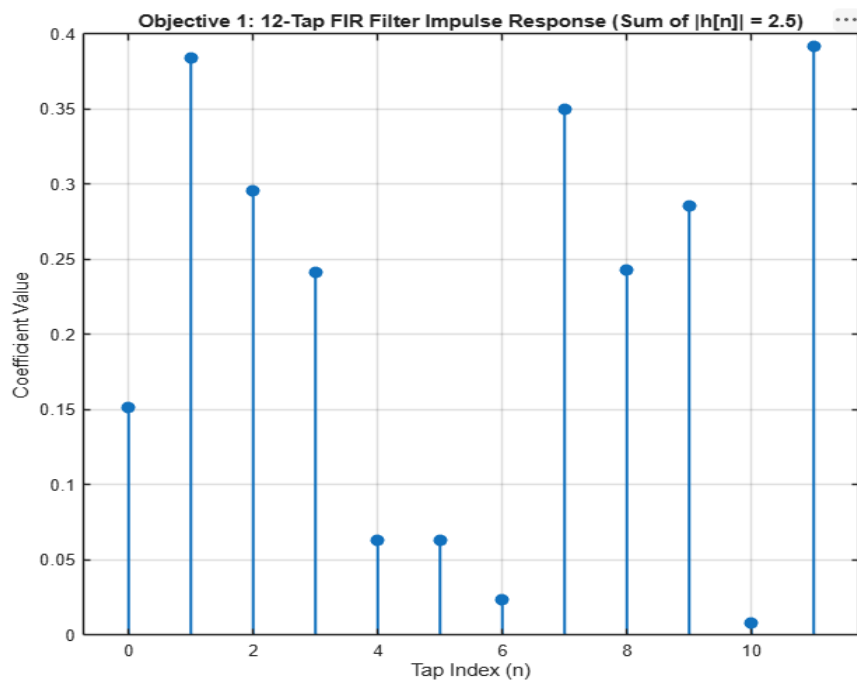
Plot a side-by-side comparative analysis between ideal, unscaled (clipped), and scaled waveforms, highlighting overflow points to demonstrate 100% overflow prevention.

MATLAB Code & Graphical Outputs

Objective 1: Scaling Factor Determination & Filter Design

/MATLAB Drive/Dsp_CO3_AT2.m

```
1 % --- Objective 1: Determine Scaling Factor and Design FIR Filter ---
2 clear; clc; close all;
3
4 % Parameters
5 N_taps = 12;           % 12 Taps
6 sum_coeff = 2.5;      % Sum of absolute coefficients
7 word_length = 8;      % 8-bit word length
8 frac_length = 7;      % Q0.7 fixed-point representation
9
10 % 1. Determine Scaling Factor
11 scaling_factor = 1 / sum_coeff; %  $S = 1 / 2.5 = 0.4$ 
12 fprintf('Objective 1: Calculated Scaling Factor S = %.2f\n', scaling_factor);
13
14 % 2. Design 12-tap FIR filter matching given criteria
15 rng(42); % Seed for consistency
16 h_raw = rand(1, N_taps);
17 h = (h_raw / sum(abs(h_raw))) * sum_coeff; % Rescale  $\sum(|h|) = 2.5$ 
18
19 % Plot Impulse Response
20 figure('Color', [1 1 1]);
21 stem(0:N_taps-1, h, 'filled', 'LineWidth', 1.5, 'MarkerSize', 6);
22 grid on;
23 title('Objective 1: 12-Tap FIR Filter Impulse Response (Sum of  $|h[n]| = 2.5$ )');
24 xlabel('Tap Index (n)');
25 ylabel('Coefficient Value');
```

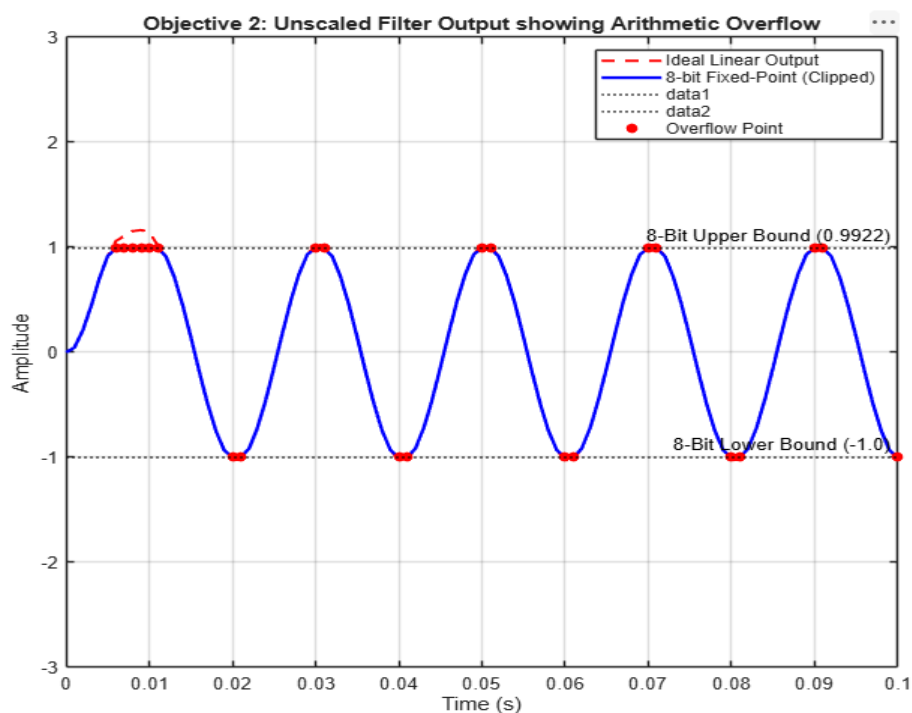


Objective 2: Unscaled Fixed-Point FIR Filter Simulation & Overflow Detection

```

1 % --- Objective 2: Unscaled Filter Simulation & Overflow Detection ---
2 % Time setup & Input signal (Amplitude +/- 1)
3 fs = 1000;
4 t = 0:1/fs:0.1;
5 x = sin(2*pi*50*t); % Input signal amplitude within [-1, 1]
6
7 % Fixed-point boundaries for signed 8-bit (Q0.7)
8 max_val = 1 - 2^(-frac_length); % 0.9922
9 min_val = -1; % -1.0000
10
11 % Filter unscaled signal
12 y_unscaled_ideal = filter(h, 1, x);
13
14 % Apply 8-bit quantization clipping (Saturation)
15 y_unscaled_fixed = y_unscaled_ideal;
16 y_unscaled_fixed(y_unscaled_ideal > max_val) = max_val;
17 y_unscaled_fixed(y_unscaled_ideal < min_val) = min_val;
18
19 % Identify Overflow Locations
20 overflow_idx = find(y_unscaled_ideal > max_val | y_unscaled_ideal < min_val);
21
22 % Plot Unscaled Simulation Results
23 figure('Color', [1 1 1]);
24 plot(t, y_unscaled_ideal, 'r--', 'LineWidth', 1.5, 'DisplayName', 'Ideal Linear Output');
25 hold on;
26 plot(t, y_unscaled_fixed, 'b-', 'LineWidth', 1.8, 'DisplayName', '8-bit Fixed-Point (Clipped)');
27 yline(max_val, 'k:', '8-Bit Upper Bound (0.9922)', 'LineWidth', 1.2);
28 yline(min_val, 'k:', '8-Bit Lower Bound (-1.0)', 'LineWidth', 1.2);
29 plot(t(overflow_idx), y_unscaled_fixed(overflow_idx), 'ro', ...
30      'MarkerSize', 5, 'MarkerFaceColor', 'r', 'DisplayName', 'Overflow Point');
31 grid on;
32 title('Objective 2: Unscaled Filter Output showing Arithmetic Overflow');
33 xlabel('Time (s)');
34 ylabel('Amplitude');
35 legend('Location', 'northeast');
36 ylim([-3, 3]);

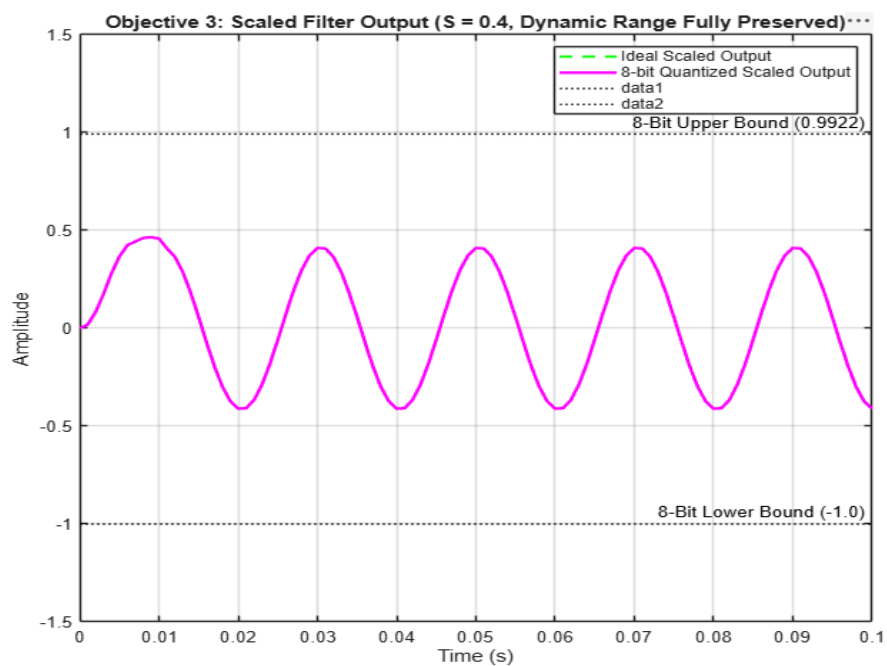
```



Objective 3: Scaled Fixed-Point FIR Filter Simulation

/MATLAB Drive/Dsp_CO3_AT2.m

```
1 % --- Objective 3: Scaled Filter Simulation ---
2 % Scale input signal by factor S = 0.4
3 x_scaled = x * scaling_factor;
4
5 % Filter scaled signal
6 y_scaled_ideal = filter(h, 1, x_scaled);
7
8 % Quantize output to 8-bit limits
9 y_scaled_fixed = y_scaled_ideal;
10 y_scaled_fixed(y_scaled_ideal > max_val) = max_val;
11 y_scaled_fixed(y_scaled_ideal < min_val) = min_val;
12
13 % Plot Scaled Simulation Results
14 figure('Color', [1 1 1]);
15 plot(t, y_scaled_ideal, 'g--', 'LineWidth', 1.5, 'DisplayName', 'Ideal Scaled Output');
16 hold on;
17 plot(t, y_scaled_fixed, 'm-', 'LineWidth', 1.8, 'DisplayName', '8-bit Quantized Scaled Output');
18 yline(max_val, 'k:', '8-Bit Upper Bound (0.9922)', 'LineWidth', 1.2);
19 yline(min_val, 'k:', '8-Bit Lower Bound (-1.0)', 'LineWidth', 1.2);
20 grid on;
21 title('Objective 3: Scaled Filter Output (S = 0.4, Dynamic Range Fully Preserved)');
22 xlabel('Time (s)');
23 ylabel('Amplitude');
24 legend('Location', 'northeast');
25 ylim([-1.5, 1.5]);
```



Objective 4: Comparative Output Analysis & Scaling Effectiveness Verification

/MATLAB Drive/Dsp_CO3_AT2.m

```
1 % --- Objective 4: Comparative Waveform Analysis ---
2 figure('Color', [1 1 1], 'Position', [100, 100, 900, 600]);
3
4 % Top Subplot: Unscaled vs Fixed-Point Limits
5 subplot(2,1,1);
6 plot(t, y_unscaled_ideal, 'r--', 'LineWidth', 1.2, 'DisplayName', 'Unscaled (Ideal)');
7 hold on;
8 plot(t, y_unscaled_fixed, 'b-', 'LineWidth', 1.5, 'DisplayName', 'Unscaled (8-Bit Clipped)');
9 plot(t(overflow_idx), y_unscaled_fixed(overflow_idx), 'ro', 'MarkerSize', 4, 'DisplayName', 'Overflows');
10 yline(max_val, 'k:'); yline(min_val, 'k:');
11 grid on;
12 title('Unscaled Filter: Severe Distortion & Overflow');
13 xlabel('Time (s)'); ylabel('Amplitude');
14 legend('Location', 'northeast');
15
16 % Bottom Subplot: Scaled vs Fixed-Point Limits
17 subplot(2,1,2);
18 plot(t, y_scaled_fixed, 'g-', 'LineWidth', 1.8, 'DisplayName', 'Scaled Output (S = 0.4)');
19 hold on;
20 yline(max_val, 'k:', '8-Bit Upper Bound');
21 yline(min_val, 'k:', '8-Bit Lower Bound');
22 grid on;
23 title('Scaled Filter: Clean Output with Zero Overflows');
24 xlabel('Time (s)'); ylabel('Amplitude');
25 legend('Location', 'northeast');
```

